DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW
SESSION 2025-2026 : WORKSHEET-1
Subject: Science Biology(086)
Chapter:5 – Our Enviornmerut

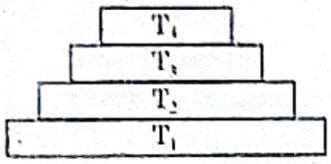
Date: _____ Name : _____ Class / Sec. X _____

Q1. The process of entry of pollutants into a food chain is known as
(a) biomass (b) biomagnification (c) bioaccumulation (d) biosphere

Q.2. In the food chain given below, if the amount of energy available at fourth trophic level is 3 kJ, what was the energy available at the herbivore level?
Grass → Grasshopper → Frog → Snake → Hawk
(a) 3000 kJ (b) 300 kJ (c) 30 kJ (d) 3 kJ

Q.3. Acid rain is caused by
(a) sulphur oxides only (b) nitrogen oxides only (c) both nitrogen and sulphur oxides (d) none of these

Q.4. In the given figure, various trophic levels are shown in a pyramid. At which trophic level minimum energy is available?



(a) T_4 (b) T_2 (c) T_1 (d) T_3

Q.5. The right path of energy flow in an ecosystem is
(a) Producer → Herbivore → Carnivore (b) Producer → Carnivore → Herbivore
(c) Herbivore → Carnivore → Producer (d) Herbivore → Producer → Carnivore.

Q.6. Minamata disease is due to
(a) MIC gas (b) methyl mercury (c) lead nitrate (d) cobalt chloride.

Q.7. According to Charles Elton, which is not correct?
(a) Carnivores at the top of the pyramid. (b) Producers at the top of the pyramid.
(c) Energy trapping is high at the top of the pyramid. (d) Both (b) and (c)

Q.8. Which of the followings is abiotic component of an ecosystem?
(a) Producer (b) Consumer (c) Soil and minerals (d) All of these

Q.9. In a food chain usually the following are found at the third trophic level .
(a) producers (b) primary consumer (c) decomposer (d) secondary consumer.

Q.10. In a given food chain if frog has 100 J of energy then the energy available with plants and snake respectively will be
Plants → Insect → Frog → Snake
(a) 1000 J and 10 J (b) 10000 J and 10 J (c) 10 J and 1000 J (d) 1000 J and 100 J.

Q11.Which of the following is biodegradable?
(a) Plastic mugs (b) Leather belts (c) Silver foil (d) Iron nails

Q12. Which of the following is non- biodegradable?
(a) Wool (b) Nylon (c) Animal bones (d) Tea leaves

Q13. Which one of the following will undergo fastest bio-degradation?
(a) Mango seed (b) Wood (c) Mango peel (d) Mango pulp

Q14

An ecosystem is represented in the figure given above. This ecosystem will be self- sustaining if
(a) the type of organisms represented by B are eliminated.
(b) materials cycle between the organisms labelled A and the organisms labelled B.
(c) the organisms labelled A outnumber the organisms labelled B.
(d) the organisms labelled A are equal in number to the organisms labelled B.

Q15. In an ecosystem, herbivores represent
(a) producers (b) primary consumers (c) secondary consumers (d) decomposers

Q16. Trophic level in an ecosystem represents
(a) oxygen level (b) water level (c) energy level (d) salt level

Q17. A food chain comprising birds, green plants, fish and man.
The concentration of harmful chemical entering the food chain will be maximum in
(a) green plants (b) man (c) birds (d) fish

Q18. First link in any food chain is usually green plants because
(a) they are widely distributed (b) they are fixed at one place in the soil
(c) they alone have the capacity to synthesise food using sunlight (d) there are more herbivores than carnivores

Q19. Which of the following statements about food chain and energy flow through ecosystem is false?
(a) Food webs include two or more food chains.
(b) All organisms that are not producers are consumers.
(c) A single organism can feed at several trophic levels.
(d) Detritivores feed at all trophic levels except the producer level.
(e) The lower the trophic level at which an organism feeds, the more energy available.

Q20. Which of the following is a logical sequence of food chain
(a) producer → consumer → decomposer (b) producer → decomposer → consumer
(c) consumer → producer → decomposer (d) decomposerproducer → consumer

Q21. Which of the following is an autotroph?
(a) Lion (b) Insect (c) Tree (d) Mushroom

Q22. In the garden ecosystem, which of the following are producers?
(a) Insects (b) Snakes (c) Grasses (d) Rabbits
Q23. Which one of the following is an artificial ecosystem? (a) Pond (b) Crop field (c) Lake (d) Forest

Q24. An ecosystem includes
(a) all living organisms (b) non-living objects
(c) both living organisms and non-living objects (d) sometimes living organisms and sometimes non living objects

Q25. Excessive exposure of humans to UV-ray s results in
(i) damage to immune system (ii) damage to lungs (iii) skin cancer (iv) peptic ulcer

(a) (i) and (ii) (b) (ii) and (iv)
(c) (ii) and (iii) (d) (iii) and (iv)

Q26. Which group of organisms are not constituents of a food chain?
(a) Grass, lion, rabbit (b) Plankton, man, fish, grasshopper
(c) Wolf, grass, snake, tiger (d) Frog, snake, eagle, grass, grasshopper

Q27. If a grasshopper is eaten by a frog, then the energy transfer will be from
(a) producer to decomposer (b) producer to primary consumer
(c) primary consumer to secondary consumer (d) secondary consumer to primary consumer

Q28. Excessive exposure to ultraviolet radiation causes
(a) inflammation of liver (b) cancer of skin (c) damage to the lungs (d) jaundice

Q29. Which one of the following is an artificial ecosystem?
(a) Lake (b) Forest (c) Pond (d) Crop field

Q30. Acid rain is caused by the oxides of
(a) carbon (b) nitrogen only (c) sulphur only (d) sulphur and nitrogen

Q31. Which of the following is biodegradable?
(a) Aluminium can (b) Polythene bag (c) Cowdung (d) DDT

Q32. Which of the following is an abiotic component of an ecosystem?
(a) Humus (b) Bacteria (c) Plants (d) Fungi

Q33. Which one of the following pairs belong to the category of primary consumers?
(a) Eagle and snake (b) Grasshoppers & cattle
(c) Snake and frog (d) Water beetles & fish

Q34. Which of the following chemicals causes depletion of the ozone layer?
(a) Carbon tetrachloride (b) Methane (c) Chloro fluoro carbon (d) Carbon monoxide

Q35. In a food chain, the third trophic level is always occupied by
(a) herbivore (b) carnivore (c) decomposer (d) producer

Q36. The depletion of the ozone layer causes
(a) global warming (b) earthquakes (c) increased UV radiations (d) acid rain

Q37. In the given foodchain if the amount of energy at the fourth trophic level is 4 kJ, what will be the energy available at the producer level?
Grass → Grasshopper → Frog → Snake
(a) 4 kJ (b) 40 kJ (c) 400 kJ (d) 4000 kJ

Q38. What will happen if all the deer are killed in the given food chain?
Grass → Deer → Lion
(a) The population of grass decreases.
(b) The population of lions increases.
(c) The population of lions remains unchanged.
(d) The population of lions decreases and grass increases.

Q39. Which of the two in the following sets belong to the same trophic level?
(a) Goats; Grasshopper (b) Goat; Spider (c) Hawk ; Rat (d) Frog ; Lizard

Direction (Q40 to Q44): In the following Questions, the Assertion and Reason have been put forward. Read the statements carefully and choose the correct alternative from the following:
(a) Both the Assertion and the Reason are correct and the Reason is the correct explanation of the Assertion.
(b) The Assertion and the Reason are correct but the Reason is not the correct explanation of the Assertion.

(c) Assertion is true but the Reason is false.
(d) The statement of the Assertion is false but the Reason is true.

Q40. Assertion: Vegetarian food habit is more beneficial to organisms.
Reason: Only 10% energy is available as food from one trophic level to next.

Q41. Assertion: Accumulation of harmful chemicals is maximum in case of organisms at higher trophic level.
Reason: Food chain normally do not go beyond 3 or 4 trophic level.

Q42. Assertion: Ozone layer is getting depleted at upper atmosphere which is a cause of concern.
Reason: CFC reacts with ozone and breaks it.

Q43. Assertion: Autotrophs can produce food on its own.
Reason: Green plants can absorb 1% energy of sunlight that fall on the leaves.

Q44. Assertion: Biodegradable waste and non biodegradable waste should be discarded separately.
Reason: Biodegradable waste are not harmful.

Q45. Direction: Match Column I with Column II.

Column I	Column II
1. Producers	(i) Suspended
2. Primary consumers	(ii) Group of colours
3. Secondary consumers	(iii) Scattering
4. Decomposers	(iv) Changing

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